

BLACK BOX TESTING - GYM MANAGEMENT SYSTEM

1) EQUIVALENCE CLASS PARTITIONING

Identified Inputs And Equivalence Classes:

Input Type	Valid Equivalence Class	Invalid Equivalence Class
Username	Non-empty, 5–15 characters	Empty, <5 chars, >15 chars
Password	8–20 characters, includes letters and numbers	Empty, <8 chars, only letters or only numbers
Age	Numeric, between 16 and 70	<16, >70, non-numeric
Email	Proper format (e.g., abc@example.com)	Blank, missing '@', missing domain
Booking Date	Future date	Past date
Payment Amount	Positive number (>0)	Zero or negative number

Test Cases Derived from Equivalence Classes:

Test Case ID	Input Field	Test Data	Input Class	Expected Output
TC1	Username	fitnessFan123	Valid	Signup success
TC2	Username	(blank)	Invalid	"Username required" error
TC3	Password	Pass1234	Valid	Login success
TC4	Password	pass	Invalid	"Password too short" error
TC5	Age	25	Valid	Registration proceeds
TC6	Age	12	Invalid	"Minimum age requirement not met"
TC7	Email	user@mail.com	Valid	Signup success
TC8	Email	usermail.com	Invalid	"Invalid email format"
TC9	Booking Date	2025-05-15	Valid	Booking confirmed
TC10	Booking Date	2023-05-01	Invalid	"Booking date cannot be in the past"
TC11	Payment Amount	5000	Valid	Payment successful
TC12	Payment Amount	-300	Invalid	"Invalid payment amount" error

Conclusion:

Using Equivalence Class Partitioning, we systematically reduced the total number of required test cases while maintaining complete coverage for functional input validation. This ensures robustness of the Gym Management System's input-handling mechanisms without the need for exhaustive testing.

2) BOUNDARY VALUE ANALYSIS

Identified Inputs for Boundary Value Analysis:

Input	Valid Range
Username Length	5–15 characters
Password Length	8–20 characters
Age	16–70 years
Payment Amount	> 0
Booking Date	Must be future date

BOUNDARY VALUE TEST CASES

Test Case ID	Input	Test Data	Expected Result
BVA1	Username Length	4 characters	Fail (too short)
BVA2	Username Length	5 characters	Pass (valid)
BVA3	Username Length	6 characters	Pass (valid)
BVA4	Username Length	14 characters	Pass (valid)
BVA5	Username Length	15 characters	Pass (valid)
BVA6	Username Length	16 characters	Fail (too long)
BVA7	Password Length	7 characters	Fail (too short)
BVA8	Password Length	8 characters	Pass (valid)
BVA9	Password Length	9 characters	Pass (valid)
BVA10	Password Length	19 characters	Pass (valid)
BVA11	Password Length	20 characters	Pass (valid)
BVA12	Password Length	21 characters	Fail (too long)
BVA13	Age	15 years	Fail (too young)
BVA14	Age	16 years	Pass (valid)
BVA15	Age	17 years	Pass (valid)
BVA16	Age	69 years	Pass (valid)
BVA17	Age	70 years	Pass (valid)
BVA18	Age	71 years	Fail (too old)

Test Case ID	Input	Test Data	Expected Result
BVA19	Payment Amount	0	Fail (invalid)
BVA20	Payment Amount	1	Pass (valid)
BVA21	Payment Amount	2	Pass (valid)
BVA22	Booking Date	Today's date (edge)	May Pass (depends)
BVA23	Booking Date	Past date	Fail (invalid)
BVA24	Booking Date	Future date	Pass (valid)